

Khang Tran

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Education

Princeton University, BA in Physics; CS & Engineering minors – Princeton, NJ Sept 2024 – May 2028

- **GPA:** 3.9/4.0
- **Relevant Coursework:** Fixed Income, Options & Derivatives / Stochastic Calculus; Financial Risk Management (Fall 2026); Honors Linear Algebra; Complex Analysis (Fall 2026); Machine Learning; Algorithms & Data Structures

NUS High School of Mathematics and Science, Diploma (High Distinction) with Honours in Jan 2020 – Oct 2023
Engineering & CS; Majors in Mathematics, Chemistry & Physics – Singapore

- **GPA:** 4.9/5.0 — Statistics, Advanced Statistics; Computer Organization, Operating Systems (NUS dual enrollment)

Honors and Competitions

- **Pyka Prize for Physics Excellence**, Princeton Physics Dept. (2026); Singapore Physics Olympiad **Gold** (2022); Singapore Junior Physics Olympiad **Gold** (2021); Singapore Physics League (SPhL) **Gold** (2022 — 8th place, 2023 — 3rd place)
- **COSCON 2025 1st** (Undergraduate); COSCON 2024 5th (Undergraduate); IBM Qiskit Hackathon **Champion** (Hardware)

Experience

Undergraduate Researcher, Computational Physics, Roberto Car Group – Princeton, NJ June 2026 – present

- Built a fault-tolerant, auto-relaunching HPC pipeline (SLURM, multi-node) running large-scale **hybrid-DFT path-integral MD** (192 atoms, 8 ring-polymer beads), auto-recovering from failures to generate model-training datasets.

Research Intern, Climate Data Analysis, High Meadows Environmental Institute – Princeton, NJ June 2025 – Aug 2025

- Built Python pipelines to analyze large multi-model time-series ensembles, quantifying how forcing strength drives a nonlinear climate response.

Space Systems Engineer Intern, Aliena – Singapore Oct 2023 – Mar 2024

- Built an end-to-end Python automation pipeline for real-time control and data acquisition during hardware ignition tests.

Projects

Statistical Arbitrage on Institutional-Holdings Asset Embeddings July 2026 – present

Independent project, advised by Prof. Caio Almeida, Princeton

- Reproduced institutional-holdings **asset embeddings** (Gabaix-Koijen-Richmond-Yogo) from free SEC 13F filings and CRSP — point-in-time universe, look-ahead-free filing-date gating, delisting-adjusted returns — then per-quarter Word2Vec embeddings across 12 quarters.
- Quantified an embedding-stability floor the source work leaves unmeasured — top-10 peer overlap of only **0.13-0.51 across seeds** — raised it to **0.75** with a 10-seed Procrustes ensemble, and showed the gain carries through to P&L.
- Backtested Avellaneda-Lee OU mean reversion on embedding k-NN peer baskets — **gross Sharpe 1.24 / net 0.37** after 10 bps costs at 22.5%/day turnover, with a scrambled-peer control (net **-11.8**) isolating cost, not signal, as the binding constraint.

Failure-Mode Analysis of Machine-Learned Interatomic Potentials May 2026

Independent study; poster, Princeton AI Fluency Lab (May 2026)

- Stress-tested **six** pretrained ML models marketed as broadly transferable against a high-accuracy reference on the simplest reaction in chemistry — errors of **5.1–72.4 kcal/mol** against a ~ 1 kcal/mol standard, all six predicting an incorrect transition state and only two recovering the reaction path when handed the correct one.

Real-Time 3D Rendering Engine (from scratch, C++ / OpenGL) July 2026 – present

- Built a real-time 3D rendering engine from scratch in **C++ 17 / OpenGL 3.3** (no engine or physics library) and benchmarked it with GPU timer queries — **188 FPS mean / 137 FPS 1% low** at 1080p, **$\sim 450K$ particles** and **2,560 rigid bodies** above 60 FPS, and **4,149 $\times$ fewer draw calls / 3.9 $\times$ faster frames** from instancing.

Leadership and Teaching

- **Event Organizer**, Princeton Science Olympiad & **Problem Setter**, SPhL; **Physics Tutor**, Princeton Physics Dept. & **CS Lab TA**, COS217/226 (2023–2026).

Skills

Programming: C/C++, Python (NumPy, Pandas, PyTorch, scikit-learn), SQL, MATLAB

Quantitative: Probability, statistics, stochastic processes, numerical methods, statistical/ML modeling